

ZIZHOU(GARY) LIU

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PUBLICATIONS

- **Liu, Zizhou**, Hui, Z., Collier, N., 2026. Persuasion Laundering in Model Ecosystems: Small-to-Large Chain Amplification of Jailbreak Compliance. (Submitted to ACM FAccT 2026).
- **Liu, Zizhou**, Gong, Z., Ai, L., Hui, Z., Chen, R., Leach, C. W., Greene, M. R., Hirschberg, J., 2025. A Review of Incorporating Psychological Theories in LLMs. (EACL 2026).
- Liu, J., Ai, L., **Liu, Zizhou**, Karisani, P., Hui, Z., Fung, M., Nakov, P., Hirschberg, J., Ji, H., 2024. PropaInsight: Toward Deeper Understanding of Propaganda in Terms of Techniques, Appeals, and Intent. (COLING 2025).
- Ai, L., Kumarage, T., Bhattacharjee, A., **Liu, Zizhou**, Hui, Z., Davinroy, M., Cook, J., Cassani, L., Trapeznikov, K., Kirchner, M., Basharat, A., Hoogs, A., Garland, J., Liu, H., Hirschberg, J., 2024. Defending Against Social Engineering Attacks in the Age of LLMs. (EMNLP 2024).
- Ai, L., Hui, Z., **Liu, Zizhou**, Hirschberg, J., 2024. Enhancing Pre-Trained Generative Language Models with Question Attended Span Extraction on Machine Reading Comprehension. (EMNLP 2024).
- Ai, L., Gupta, S., Oak, S., Hui, Z., **Liu, Zizhou**, Hirschberg, J., 2024. TweetIntent@Crisis: A Dataset Revealing Narratives of Both Sides in the Russia-Ukraine Crisis. In: Proceedings of the International AAAI Conference on Web and Social Media. (AAAI ICWSM2024)
- Ai, L., **Liu, Zizhou**, Hirschberg, J., 2022. Combating the COVID-19 Infodemic: Untrustworthy Tweet Classification using Heterogeneous Graph Transformer. (Mediate@AAAI ICWSM2023)
- Cao, I.*, **Liu, Zizhou***, Karamanolakis, G., Hsu, D., Gravano, L., 2021. Quantifying the effects of COVID-19 on restaurant reviews. In: Proceedings of the Ninth International Workshop on Natural Language Processing for Social Media (SocialNLP@NAACL 2021)

EDUCATION

Columbia University <i>Master of Science in Computer Science, NLP track</i>	Graduated Feb 2023
Columbia University <i>Bachelor of Science in Computer Science, Intelligent Systems track</i>	Graduated May 2021

RESEARCH EXPERIENCE

LLM Alignment Projects Advised by Prof. Nigel Collier <i>Research Assistant</i>	Cambridge University Aug 2025
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- PROJECT: Persuasion Laundering in LLM Ecosystems: Small-to-Large Chain Amplification of Jailbreak Compliance
 - ◊ Developed a chain-based red-teaming framework demonstrating small-to-large cascade amplification of jailbreak outcomes in reasoning-model ecosystems .
 - ◊ Introduced “persuasion laundering” as a mechanistic lens—grounded in credibility/trust propagation—explaining how intermediate models can increase downstream susceptibility without changing the underlying intent.
 - ◊ Flagged systemic “weakest-link” risk in coupled model pipelines: localized unsafe behavior in any component (including 0.6B reasoning-capable SLMs) can trigger ecosystem-wide safety degradation via cascading.

Semantic Forensics(SemaFor) Advised by Prof. Julia Hirschberg <i>Research Associate</i>	Columbia University May 2022 - Aug 2025
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- PROJECT: A Survey of Incorporating Psychological Theories in LLMs
 - ◊ Led an interdisciplinary survey mapping where psychological theories are used in the LLM lifecycle (preprocessing, pre-training, post-training, evaluation and application).
 - ◊ Identified patterns of conceptual drift in how AI research adopts constructs like Theory of Mind, attention and memory, and proposed guidelines for theory-faithful model to support psychologically grounded alignment.
- PROJECT: Defending Against Social Engineering Attacks in the Age of LLMs
 - ◊ Designed the ConvoSentinel pipeline, a modular defense system that enhances detection of social engineering (SE) attacks at both message and conversation levels, incorporating retrieval-augmented generation (RAG) to effectively identify malicious intent with improved accuracy and adaptability.
 - ◊ Achieved a 12% F1 score improvement over baseline models in SE detection by integrating message-level SI analysis and snippet-based malicious intent identification, significantly enhancing early-stage detection robustness.
- PROJECT: QASE Enhanced PLMs: Improved Control in Text Generation for MRC
 - ◊ Developed the Question-Attended Span Extraction (QASE) module, enhancing pre-trained language models (PLMs) like Llama2, Alpaca and FlanT5 during fine-tuning for machine reading comprehension (MRC).

◇ Achieved performance comparable to SOTA models designed for MRC tasks and surpassing leading LLMs like GPT-4 without substantial increases in computational costs.

Social Media and Public Health Advised by Prof. Luis Gravano & Prof. Daniel Hsu Columbia University
Research Assistant *Aug 2020 - Aug 2022*

- PROJECT: Effect of Verbalizers on Prompt-based Few Shot Learning
 - ◇ Analyzed impact of different verbalizers in prompt on BERT and RoBERTa model in a zero/few-shot setting.
 - ◇ Proposed Soft-verbalizer(version of soft prompt) that merge embeddings of verbalizers and improves the accuracy
- PROJECT: Quantifying the Effects of COVID-19 on Restaurant Reviews
 - ◇ Evaluated different COVID-19 aspect extraction(BERT, Student-Teacher model and Bag of Words).
 - ◇ Conducted time-series analysis on Yelp reviews and critical COVID-19 statistics.

PROFESSIONAL EXPERIENCE

Loxo Talent Intelligence Platform Remote
Machine Learning Engineer *April 2024 - present*

- Finetuned, designed and accelerated many models/pipelines at different scale(0.3 to 13b)(Qwen family, Llama family, Phi family, BERT family and API-based LLMs) for experience merging, title resolution, resume matching, profile ranking, email outreach agent with RLHF, RAG, LoRA etc.

Columbia University Remote
Staff Research Associate, Conversational AI Lab *Oct 2023 - April 2024*

- Utilized OpenAI, Llama, and Langchain with dialog-flow, tree of thoughts, intent characterization and retrieval augmented generation to build SOTA chatbot that drive sales and education engagement.
- Chatbot examples can be found at <https://www.articulateai.com/>, <https://www.richtechrobotics.com/>, etc.

Black Sesame Technologies San Jose
Machine Learning Engineer *Jan 2023 - Oct 2023*

- Improved inference latency and power consumption of DNNs by 2 to 3 times through magnitude pruning, dynamic step pruning, int-4/8 quantization and equivalent layer/node/model structure substitution.